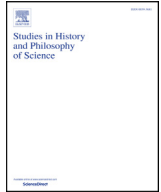




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Saving the mutual manipulability account of constitutive relevance

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ABSTRACT

Constitutive mechanistic explanations are said to refer to mechanisms that *constitute* the phenomenon-to-be-explained. The most prominent approach of how to understand this relation is Carl Craver's mutual manipulability approach (MM) to constitutive relevance. Recently, MM has come under attack (Baumgartner and Casini 2017; Baumgartner and Gebharder 2015; Harinen 2014; Kästner 2017; Leuridan 2012; Romero 2015). It is argued that MM is inconsistent because, roughly, it is spelled out in terms of interventionism (which is an approach to causation), whereas constitutive relevance is said to be a non-causal relation. In this paper, I will discuss a strategy of how to resolve this inconsistency—so-called *fat-handedness approaches* (Baumgartner and Casini 2017; Baumgartner and Gebharder 2015; Romero 2015). I will argue that these approaches are problematic. I will present a novel suggestion for how to consistently define constitutive relevance in terms of interventionism. My approach is based on a causal interpretation of manipulability in terms of causal relations between the mechanism's components and what I will call *temporal EIO-parts* of the phenomenon. Still, this interpretation accounts for the fundamental difference between constitutive relevance and causal relevance.

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1. Introduction

Defenders of the new mechanistic approach highlight the importance of so-called *constitutive* mechanistic explanations for the life sciences and other special sciences. Constitutive mechanistic explanations are taken to explain a phenomenon in terms of a mechanism that *constitutes* the phenomenon (Bechtel, 2008; Bechtel & Abrahamsen, 2005; Craver, 2007b; Craver & Darden, 2013; Illari and Williamson 2012; Machamer, Darden, and Craver 2000). A standard view is that this constitution-relation is a *non-causal dependency relation* that holds between parts (the mechanism's components) and a whole (the phenomenon). According to Carl Craver's popular mutual manipulability approach (MM), this non-causal dependency relation is to be analyzed in terms of *mutual manipulability* between the phenomenon's parts and the phenomenon (Craver, 2007b, p. 153), where mutual manipulability is spelled out in terms of Woodwardian *interventionism* (Woodward, 2003).

At a first glance, MM captures many aspects of the explanatory and experimental practice of the life sciences, especially the practice of interlevel experiments (Craver, 2002; Kaplan, 2012; Romero, 2015). Still, recently different authors have argued that MM is problematic because of its attempt to spell out a non-causal

dependency relation in terms of interventionism (Baumgartner & Casini, 2017; Baumgartner & Gebharder, 2015; Harinen, 2014; Kästner, 2017; Leuridan, 2012; Romero, 2015). This is problematic because interventionism draws on the notion of an *ideal* intervention. Ideal interventions have to satisfy what may be called the *transmission condition*¹: in order for an intervention *I* into a variable *X* with respect to some other variable *Y* to be ideal *I* must change *Y* only indirectly via *X*. Michael Baumgartner and Alexander Gebharder (2015), Baumgartner and Lorenzo Casini (2017), Felipe Romero (2015), and Lena Kästner (2017) show that this condition is violated by interventions into phenomena with respect to their mechanistic components. These variables are necessarily *fat-handed*,² i.e., they change the component-variable on a causal path that does not go through the phenomenon-variable. As a consequence, ideal interventions into phenomena that are constituted by mechanisms are impossible. Hence, interventionism cannot be applied to constitutive relations. Therefore, MM fails.

There are different ways to react to this problem. One way is to simply reject the whole endeavor of trying to spell out constitution in term of interventionism. For example, Couch (2011) and

¹ There are further conditions; see Woodward (2003, 89), and Craver (2007b, 154).

² Woodward characterizes interventions as fat-handed if they affect "not just *X* and other variables lying on the route from *I* to *X* to *Y*, but also other variables that are not on this route and that affect *Y*" (Woodward, 2008, p. 209).

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Harbecke (2010) defend a regularity account of mechanistic constitution. Gillett (2013) argues that his dimensioned realization approach best accounts for mechanistic constitution. Gebharter (2016) investigates to what extent the PC algorithm, that was originally developed for detecting causal relations, can be used to discover constitutive relevance relations. Kästner (2017) suggests non-interventionist research strategies to distinguish causal from non-causal dependency relations. Another strategy is to interpret constitutive explanation as a variant of causal explanation (Harinen, 2014; Leuridan, 2012), and thereby reject the fundamental difference between causation and constitution and between causal and constitutive explanation.

In this paper, I want to focus on a rescue strategy that maintains the idea of using interventionism to spell out constitution but that respects the fundamental difference between causation and constitution. It has been suggested by Baumgartner and Gebharter (2015), Baumgartner and Casini (2017), and (independently) by Romero (2015). These authors exploit the notion of a *fat-handed intervention* for the purpose of providing a positive account of constitution. Both accounts rely on different formulations of a *fat-handedness criterion* that interventions have to satisfy in order to allow for inferences to constitution. In this paper, I want to analyze the two *fat-handedness approaches* and their differences. I will argue that both approaches fail for different reasons. Instead, I will provide a novel rescue strategy for MM. According to the account I will introduce here, mutual manipulability can be analyzed in terms of causal relations between a mechanistic component and what I will call *temporal EIO-parts* of the constituted phenomenon. Still, the account maintains the fundamental distinction between constitution and causation.

The paper proceeds as follows: in Section 2, I will introduce the general idea of constitutive mechanistic explanation and Craver's mutual manipulability account. In Section 3, I will present the problems that arise when combining constitutive relevance and interventionism in more detail. I will argue that in order to solve the problems two challenges have to be met. First, one has to provide a notion of an intervention that allows for inferences to causation without rendering constitutive relations causal. Second, one has to show how constitutive relevance can be spelled out in terms of interventionism. In Section 4, I will discuss different strategies for meeting the first challenge. In Section 5, I will present answers to the second challenge. I will first (5.1) present and criticize approaches based on fat-handedness. Then (5.2), I will present my own account. Section 6 concludes.

2. Constitutive explanations, mechanisms, and mutual manipulability

Different characterizations of mechanisms have been suggested. Despite their differences, one common assumption is that mechanisms consist of entities/parts/objects and their activities/interactions/operations in a certain organization (Machamer, Darden, and Craver 2000; Craver, 2007b; Illari and Williamson 2012; Craver & Darden, 2013; Glennan, 2017). Based on this characterization, usually two kinds of mechanistic explanations are distinguished (Craver, 2007b; Salmon, 1984): first, in *etiological* mechanistic explanations, a phenomenon is explained by the mechanism that *causes* it. Second, in *constitutive* mechanistic explanations, a phenomenon is explained by the *underlying* mechanism. Here, mechanism and phenomenon are not related by causation. Rather, the mechanism is taken to *constitute* the phenomenon.

One prominent example of a constitutive mechanistic explanation is the explanation of spatial memory. Spatial memory is often investigated by observing mice navigating the Morris water maze (a pool filled with an opaque liquid; the mouse is supposed to

find a platform that is hidden under the surface of the liquid). Spatial memory is usually described as being instantiated in the mouse's navigating the Morris water maze (the phenomenon), and the mouse's hippocampus generating spatial maps is supposed to be a component of the mechanism responsible for the navigation behavior (Bechtel, 2008; Bechtel & Richardson, 2010; Craver, 2007b). Other examples of phenomena that are constitutively explained are the action potential (Craver, 2007b, pp. 114–22), the human heart pumping blood (Bechtel & Abrahamsen, 2005, p. 425; Bechtel, 2006, pp. 29–30; Glennan, 2010, p. 257; Craver & Darden, 2013, pp. 98–117), a cell synthesizing proteins (Machamer, Darden, and Craver 2000; Darden, 2002; Craver & Darden, 2013), and long-term potentiation at synapses of neurons (Machamer, Darden, and Craver 2000, 8–11; Craver & Darden, 2001, 2013, pp. 115–17, pp. 167–72; Craver, 2007b, pp. 65–72) (see also Kaiser and Krickel (2016)).

What exactly does it mean to *constitutively* explain a phenomenon? When does a mechanism *constitute* a phenomenon? Craver illustrates the notion of this constitution-relation with the help of the following figure (Fig. 1):

According to Craver, “S’s Ψ -ing is explained by the organization of entities $\{X_1, X_2, \dots, X_i\}$ and activities $\{\Phi_1, \Phi_2, \dots, \Phi_n\}$ ” (Craver, 2007b, p. 7). “S” refers to the mechanism as a whole; Ψ is the behavior of S that is to be explained. The Xes represent the entities that are components of the mechanism, and the Φ s are the activities performed by the entities. The arrows stand for the different interactions between the Xes’ Φ -ings. Although the picture alone cannot provide a full understanding of what constitutive explanations are, it provides us with some information: first, the phenomenon and the mechanism occur at the same time (indicated by the fact that the phenomenon is located above the mechanism). Second, the relation between the mechanism and the phenomenon is not causal (indicated by the dotted lines; causal relations are represented as arrows). Both assumptions are commonly accepted among the new mechanists (Baumgartner & Casini, 2017; Baumgartner & Gebharter, 2015; Craver, 2007b; Leuridan, 2012; Romero, 2015).

What exactly is the relation between the mechanism/the mechanism's components and the phenomenon, if it is not causal? According to Craver (2007b; 2007a), in constitutive mechanistic explanations one refers to components of mechanisms that are *constitutively relevant* for the phenomenon. Constitutive relevance,

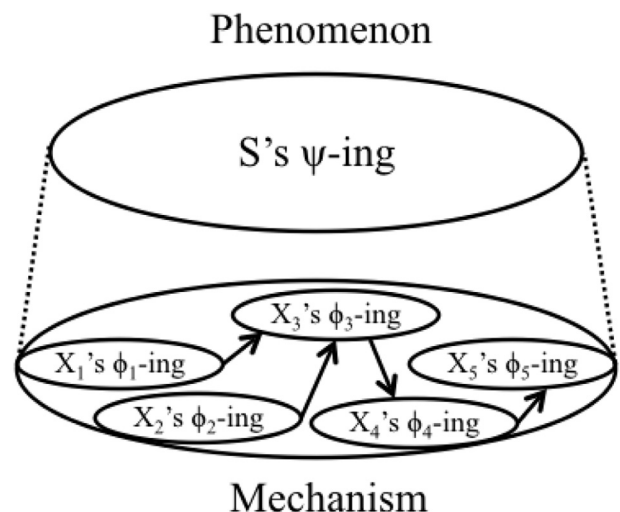


Fig. 1. Mechanisms that constitute phenomena according to Craver (adapted from Craver, 2007a, p. 7).

in contrast to constitution which holds between the phenomenon and the mechanism as a whole, is taken to be a relation that holds between the phenomenon and a component of the mechanism. Craver defines constitutive relevance as follows:

(MM) X 's Φ -ing is constitutively relevant for S 's Ψ -ing iff:

- (i) X 's Φ -ing is a part of S 's Ψ -ing,³ and
- (ii) X 's Φ -ing and S 's Ψ -ing are mutually manipulable (Craver, 2007b, p. 153).

This definition of constitutive relevance provides us with some information about what it means for a mechanism to constitute a phenomenon: first, the components of the mechanism are *spatio-temporal parts* of the phenomenon. For example, the hippocampus is a spatial part of the mouse, and the hippocampus's activity occurs during the mouse's navigation behavior. Second, constitutive relations involve *mutual manipulability*, which is taken to consist of two manipulation steps:

(Mutual Manipulability) X 's Φ -ing and S 's Ψ -ing are mutually manipulable iff:

- (i) there is an ideal intervention on X 's Φ -ing with respect to S 's Ψ -ing that changes S 's Ψ -ing;
- (ii) there is an ideal intervention on S 's Ψ -ing with respect to X 's Φ -ing that changes X 's Φ -ing (Craver, 2007b, p. 153).

Ideal interventions, thereby, are spelled out analogously to Woodwardian interventionism (Woodward, 2003; Craver, 2007b, p. 154). Roughly, an intervention I on X 's Φ -ing with respect to S 's Ψ -ing is ideal if and only if X 's Φ -ing changes only due to the influence of I and S 's Ψ -ing is not changed by I directly (I will say more on this in the next section).

Besides these two features that are explicitly stated in MM, constitutive relevance, according to Craver, has the following further features: first, as already mentioned, constitutive relevance is a non-causal relation. If an X 's Φ -ing is constitutively relevant for S 's Ψ -ing, then it is impossible that X 's Φ -ing causes S 's Ψ -ing (Craver, 2007b; Craver & Bechtel, 2007). Most arguments in favor of this claim are based on the fact that X 's Φ -ing and S 's Ψ -ing are not wholly distinct because X 's Φ -ing occupies a sub-region of the spatiotemporal region of S 's Ψ -ing (condition i.) and both are

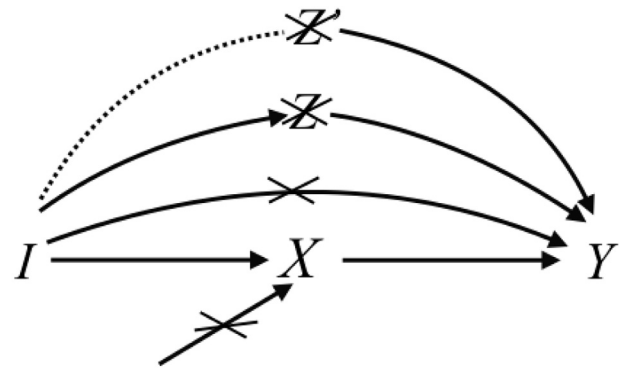


Fig. 2. Ideal intervention I on X with respect to Y .

mutually dependent (condition ii.) (Craver & Bechtel, 2007; Krickel, 2017; Romero, 2015).⁴ One fundamental assumption about causation is that its relata are wholly distinct (Craver & Bechtel, 2007; Lewis, 1973; 1986). Second, constitutive relevance (and mechanistic constitution) are not relations of strong, or “more robust kinds of” emergence (Craver, 2015). The properties of the phenomenon are determined by the properties of the components and it is impossible to change the properties of the phenomenon without changing the properties of at least some of the components. Third, constitution is compatible with multiple realization (or, rather, multiple constitution) in the sense that there might be two different mechanism types that constitute the same phenomenon type, and there may be two different components that can be interchangeably involved in the same mechanism. As a consequence, there are changes in mechanistic components that are not accompanied by changes in the phenomenon.

3. The incompatibility of constitutive relevance and interventionism

As Leuridan (2012), Baumgartner and Gebharder (2015), Romero (2015), Baumgartner and Casini (2017), and Kästner (2017) argue, there is a problem for Craver's mutual manipulability account: if one wants to uphold the intuition that constitutive relevance is not a causal relation, constitutive relevance relations are incompatible with interventionism. To see this, let me introduce the most central notions of the interventionist framework.

First, note that interventionists use variables to model causal structures. Interventionism states that a variable X is a (direct) cause of a variable Y if and only if there is an *ideal intervention* on X with respect to Y that changes Y given that all variables that are not on the causal path between I , X , and Y are held fixed (Woodward, 2003). An intervention I on X with respect to Y is ideal if and only if it satisfies various conditions—most importantly the *transmission condition*: I does not change Y directly without this change being transmitted via X . (Further conditions are: X changes only due to changes in I ; there is no further variable Z that is not on the causal path between I and X that is changed by I and influences Y ; and I is not probabilistically correlated with a further variable Z' that causes Y via a path not going through X ; see Fig. 2 for an illustration of an ideal intervention).

MM requires there to be ideal interventions on the phenomenon-variable with respect to each component-variable, and ideal interventions on each component-variable with respect to the phenomenon-variable (see Fig. 3). I use variables Φ_1 – Φ_3 to represent the mechanism's components and variable Ψ to represent the phenomenon. The interventions I_{Φ_1} , I_{Φ_2} and I_{Φ_3} are interventions on the respective component-variable with respect to

³ Craver's original definition of constitutive relevance requires only X to be a part of S (condition (i)) (Craver, 2007b, p. 153). But it is plausible to assume that constitutive relevance does not only require the component-entities to be spatial parts of the phenomenon-entity. Additionally, the behavior that is to be explained and the behavior of the components that explain the phenomenon have to occur at the same time (Baumgartner & Gebharder, 2015; Krickel, 2017; Leuridan, 2012).

⁴ A further argument in favor of the claim that constitutive relevance is a non-causal relation is that otherwise we are confronted with causal loops (Baumgartner & Gebharder, 2015; Craver & Bechtel, 2007; Romero, 2015). Causal loops are held to be problematic because, roughly, their existence would imply that something could be the cause of its own cause (Kim, 1999; Romero, 2015). Causal loops are unproblematic if they are taken to be feedback loops where an effect is a cause of an effect that is of the same type as the cause of the first effect. The problematic element comes in when assuming that the putative cause and effect occur at the same time. A further argument Romero provides is based on exclusion worries arguing that if constitutive relevance would be a causal notion that would imply that there is downward causation, which is not compatible with the causal closure of the physical. This argument does not depend on the assumption that the relata of constitutive relevance are parts and wholes. I will not discuss this objection in the present paper. Surely, given that I will argue that constitutive relevance can be spelled out in causal terms and given that this implies that there is downward-causation, exclusion worries have to be addressed. For a discussion of exclusion worries in the context of interventionism see Baumgartner (2009), Woodward (2015), and Gebharder (2015). For a discussion of exclusion worries concerning causation between mechanistic levels, see Craver (2007b, chap. 6), and between parts and wholes, see Kim (1998, 84).

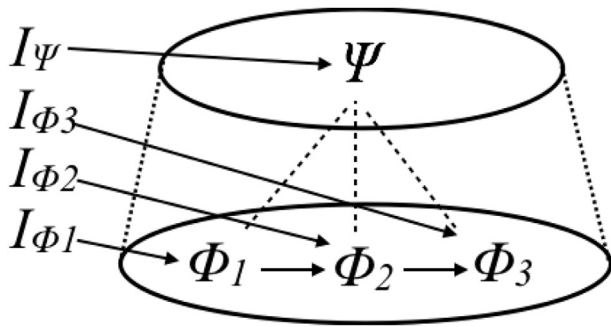


Fig. 3. Mutual manipulability requires there to be ideal interventions on the phenomenon-variable Ψ with respect to every component-variable and ideal interventions on every component-variable with respect to the phenomenon-variable (adopted from Baumgartner and Gebharder (2015, p. 741)).

the phenomenon; I_Ψ indicates that there must be interventions on the phenomenon-variable with respect to each component-variable.

The first problem is that it is impossible to ideally intervene into the phenomenon-variable Ψ because these interventions violate the transmission condition. Any intervention on Ψ is a fat-handed intervention into one of the components. To see that, assume that Φ_1 is constitutively relevant for Ψ and that there is a correlation between changes in Φ_1 and Ψ when an intervention I_Ψ on Ψ is performed. Now, according to interventionism (and Reichenbach's common cause assumption; see Romero (2015, 3745)), this correlation is either due to the fact that Ψ causes Φ_1 , or that Φ_1 causes Ψ , or that I is a common cause of Φ_1 and Ψ (for a more detailed presentation of this argument see Baumgartner and Gebharder (2015, pp. 741–42), and Kästner (2017, chap. 6)). Since constitutive relevance is supposed to be a non-causal relation, the only possible explanation for the correlation between Φ_1 and Ψ is that I is a common cause of Φ_1 and Ψ . Hence, there are no ideal interventions into phenomena that are constituted by mechanisms.

Baumgartner and Gebharder (2015, p. 745) argue that this problem can be solved by introducing Woodward's modified definition of an ideal intervention (Woodward (2015, 333); I will talk about *ideal** interventions). Roughly, Woodward argues that an intervention I on X with respect to Y is *ideal** iff I does not influence any variable that is not on the causal path between I and X except for variables that X non-causally depends on (by, e.g., definitional dependence, supervenience, realization, or constitution). Fig. 4

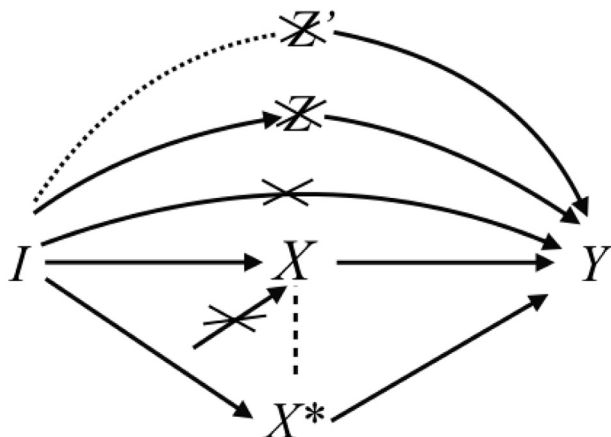


Fig. 4. *Ideal** interventions according to Woodward (2015); *ideal** interventions can be common causes of two variables that are related by a non-causal dependency relation.

illustrates an *ideal** intervention (the dotted line between X and X^* indicates that there is a non-causal dependency relation between them).

This strategy renders *ideal** interventions into mechanisms possible because *ideal** interventions can be fat-handed common causes of variables that are non-causally related.

Still, this strategy gives rise to a second problem (Baumgartner & Gebharder, 2015, p. 747). Based on the definition of *ideal** interventions, constitutive relevance turns out to be causal. To see this, note that Woodward does not only modify the notion of an ideal intervention but also the definition of causation. The modified interventionist account of causation states that X is a (direct) cause of Y iff there is an ideal or *ideal** intervention I on X with respect to Y that changes Y , and all other variables that are not on the causal path between I , X , and Y are kept fixed *except for variables that I , X , and Y non-causally depend on*. Now, it turns out that according to this modified account, the existence of an *ideal** intervention on Ψ with respect to a component-variable Φ_i establishes that Ψ is a cause of Φ_i . This is due to the fact that the modified account of interventionist causation is compatible with the fact that the other component-variables change if we ideally* intervene on Ψ with respect to Φ_i . Since Ψ non-causally depends on all component-variables, the modified account of interventionist causation states that we need not keep them fixed. Hence, the two necessary and together sufficient conditions for causation are satisfied for interventions on the phenomenon with respect to its components. This contradicts the assumption that constitutive relevance is a non-causal relation (see Section 2).

In order to avoid these two problems, we have to meet two challenges: first, we have to find a way to formulate interventionism such that it does not render constitutive relevance causal. Second, we have to provide a definition of constitutive relevance in interventionist terms. I will address both challenges in the next two sections.

4. First challenge: keeping apart causation and constitution

In order to solve the problems presented in Section 3, the first challenge is to find a definition of interventionism that does not render constitutive relevance causal.

Romero (2015) addresses this challenge by simply rejecting Woodward's modification of interventionism. For Romero, interventions into phenomena constituted by mechanisms are simply not ideal and, thus, do not establish causal relations. This strategy is not attractive because it implies that ideal interventions are almost never possible. The reason is that if physicalism is correct, every higher-level phenomenon supervenes on the physical. This implies that one cannot ideally intervene into higher-level phenomena in general. Ideal interventions would be possible only with regard to the fundamental physical. As a consequence, the scope of interventionism would be rather restricted.

Kästner (2017) accepts a modified version of interventionism similar to Woodward's but argues that, instead of being an approach to causation, interventionism* only provides a method for detecting dependency relations in general—without enabling us to say more about the nature of this dependency relation. Kästner's approach implies that we have to reject interventionism as an approach to causation. This consequence is in conflict with the fact that interventionism has proven fruitful as an account of causation. Furthermore, interventionism is taken to be a descriptively adequate (though idealizing) account of how causal discovery proceeds. If Kästner is right, we are wrong in assuming that we can test for causation by means of (ideal) intervention experiments. This, of course, does not prove that Kästner is wrong. But it provides a motivation to look for an alternative.

Baumgartner and Gebharder (2015) address the first challenge by introducing time into the interventionist definition of a cause: an ideal* intervention I on X with respect to Y that changes Y indicates a causal relationship between X and Y only if the change in X occurs before the change in Y . In contrast to that, they argue, constitution implies a simultaneous change in the two variables.

A similar strategy is to allow for an inference to causation only if the (values of the) variables represent events that are wholly distinct (which is usually an implicit requirement for causal models). Hausman and Woodward, 1999 argue that

[w]hen variables bear conceptual or logical connections to one another, or when their located values have parts in common, then they may bear probabilistic relations to one another that have no causal explanation. (523)

They specify in a footnote that

[t]oken causal relations obtain among distinct events; that is, among distinct instantiations of properties at particular spatio-temporal locations or among spatio-temporally distinct located values of variables. (523, fn. 4)

These quotes show that the assumption that the variables in a causal model have to represent wholly distinct events was already in the background of interventionism. Two events are wholly distinct only if they do not logically, definitionally, or metaphysically depend on each other (Lewis, 1973; 1986). For events to be wholly distinct they have to occupy distinct space-time regions or they do not depend on each other. The relations of constitution are not wholly distinct because they occupy the same space-time region (or rather, one occupies a sub-region of the space-time region of the other) and they are mutually dependent. Now, one could argue that an ideal* intervention I on X with respect to Y that changes Y indicates causation only if all values of I , X , and Y represent events that are wholly distinct. Since the values of variables that are constitutively related are not wholly distinct, ideal* interventions on these variables do not indicate that the variable is a cause of the other.

I will accept this last solution for the purposes of the present paper: an ideal intervention I into X with respect to Y establishes a causal relationship between X and Y only if the values of I , X , and Y represent wholly distinct events. Baumgartner and Gebharder's solution is similar since the requirement that cause and effect are wholly distinct events is satisfied if causation takes time. But the distinctness requirement is less specific because it leaves open whether causes precede their effects, occur afterwards, or occur simultaneously.

In the following section, I will discuss two approaches to meeting the second challenge: how to define constitutive relevance in terms of interventionism.

5. Second challenge: constitutive relevance in terms of interventionism

5.1. Approaches based on fat-handedness

We still need a positive account of constitutive relevance in terms of interventionism. Baumgartner and Gebharder (2015) and Romero (2015) develop approaches that make use of the fact that interventions on mechanistic phenomena are always fat-handed common causes. What has to be shown is how common causes that are due to constitutive relevance between Φ and Ψ differ from accidental common cause structures and how this difference can be

expressed in terms of interventionism. For example, imagine you intervene on a driving car by smashing it with a huge hammer, and thereby you stop the car and you destroy the windshield. Your intervention is a common cause of the car's stopping and the destruction of the windshield. But this does not show that the latter is constitutively relevant for the former. Rather, it is what I call an *accidental common cause*. Fat-handedness approaches rely on the idea that common cause structures that are due to a constitutive relevance relation differ from accidental common cause structures in that in the former case every intervention is fat-handed (call this the *fat-handedness criterion*), while in the latter case only some interventions are fat-handed. As I will show, it is unclear how to spell out this idea in detail in order to make sense of the difference between the hammer smashing example mentioned above, and cases that involve constitutive relevance relations.

According to Baumgartner and Gebharder's approach, constitution relations satisfy the fat-handedness criterion which implies that every ideal* intervention into the phenomenon is a fat-handed intervention on the phenomenon and at least one of its components (Baumgartner & Gebharder, 2015, p. 752). With respect to non-constituents, only some ideal* interventions are common causes. Although this version of the fat-handedness criterion is plausibly necessary for constitution, it does not provide us with any resources to exclude the windshield from being constitutively relevant to the car's driving. The reason is that the account is compatible with there being components that are manipulable by means of only one fat-handed intervention. Hence, Baumgartner and Gebharder's characterization of the fat-handedness criterion fails to account for the difference between accidental common causes and those that are due to constitutive relevance.

In contrast to Baumgartner and Gebharder, Romero characterizes the fat-handedness criterion in terms of individual components. Romero suggests the following working definition of constitutive relevance in terms of fat-handed interventions (Romero, 2015, p. 3746; my reconstruction)⁵:

(*Fat-handed CR*) A variable Φ_i of a variable set $V = \{\Phi_1, \dots, \Phi_n\}$ is constitutively relevant for a variable Ψ iff:

- (i) Φ_i represents a spatiotemporal part of what is represented by Ψ ,
- (ii) every intervention on Ψ is necessarily a fat-handed intervention on Ψ and one of the variables of set V ,⁶
- (iii) every intervention on Φ_i is necessarily a fat-handed intervention on Φ_i and Ψ .

The first condition corresponds to the first condition of Craver's mutual manipulability approach. The second and third conditions

⁵ Romero argues that *Fat-handed CR* provides only a necessary condition for constitutive relevance because "[a]ny time we have a fat-handed intervention between two variables we cannot infer that we are talking about a constitutive relevance relation in a mechanism (...) some fat-handed interventions are so because they aren't refined enough to have a localized effect in one variable only" (Romero, 2015, p. 3748). This comment is confusing because his definition states requirements concerning every intervention. Surely, it is an open question how we can empirically test for constitutive relevance based on Romero's approach. But the approach seems to exclude the consequence that accidentally fat-handed interventions turn out to establish constitutive relevance because in these cases we can expect to find at least one intervention that is not fat-handed. Hence, it is not clear why the account should provide only necessary conditions.

⁶ In his definition, Romero does not explicitly mention that the intervention influences "one of the variables of set V ." But since he mentions it later in his text and since it seems to be more plausible than requiring that each intervention into the phenomenon has to be an intervention into one particular component, I reconstruct his approach in this way.

constitute a reinterpretation of Craver's mutual manipulability condition.

Fat-handed CR is problematic. The first reason is that it is not clear how the “necessarily” should be interpreted given the interventionist framework. It is not clear what the “necessarily” adds beyond the assumption that every intervention is fat-handed. According to the very idea of fat-handedness approaches to constitutive relevance, invoking a fat-handedness requirement that applies to every intervention of a certain kind (as formulated in (ii) and (iii)) just means to spell out what it means for a relation to hold necessarily in interventionist terms. Hence, adding “necessarily” seems to be redundant. Of course, this may not be a big problem for Romero's approach. But it is misleading at least.

The second problem is that, from an empirical perspective, the question of whether constitutive relevance holds in a given case turns out to be a non-definite issue. In order to find constitutive relevance, one has to establish that every intervention of a certain kind is fat-handed. From an empirical perspective, it is impossible to give a definite answer. The only thing one can do is to test as many interventions as possible and if they turn out to be fat-handed one can abductively infer that this is due to the fact that the variables are constitutively related (Baumgartner & Casini, 2017). It is questionable whether scientists really make these abductive inferences. Even worse, the fact that single fat-handed interventions are always underdetermined with regard to inferences to the underlying structure plus the fact that there is always a simple, purely causal interpretation (in terms of common causes) might raise doubts with regard to whether there is sufficient justification for the whole enterprise of looking for an extra dependency relation, i.e., constitutive relevance.⁷

The third reason why *Fat-handed CR* is problematic is that it does not allow for multiple realization. Multiple realization requires there to be interventions on components that do not change the phenomenon. This requires there to be interventions onto components that are not fat-handed interventions into a component and the phenomenon. Thus, condition (iii) is too strong. But if there can be constitutively relevant parts that can be surgically intervened on (i.e., not by a fat-handed intervention), and we, thus, have to reject condition (iii), *Fat-handed CR* fails as an account of constitutive relevance. (Indeed, Baumgartner and Gebharder's approach accounts for multiple realization because they do not invoke a condition analogous to (iii).) The problem cannot be solved by simply dropping condition (iii). Condition (ii) is silent with regard to particular single components but allows for inferences only with regard to sets of variables. Romero's approach would collapse into Baumgartner and Gebharder's account—which, as mentioned above, is useless when it comes to specifying how to identify individual constituents independently of a complete set of constitutively relevant variables.

A fourth problem of the fat-handedness approach is that it cannot distinguish between top-down and bottom-up interventions since all interventions are always on both levels at the same time. This is problematic because it defeats central motivations of the original mutual manipulability account, and of the whole endeavor of combining constitutive relevance with interventionism. First, Craver introduced the notions of top-down and bottom-up interventions to make sense of different research strategies in the life sciences. According to Craver, interference experiments, stimulation experiments, and activation experiments differ in whether they are so-called *bottom-up experiments* (performing bottom-up interventions) or top-down experiments (performing

top-down interventions) (Craver, 2007b, pp. 146–47). Second, spelling out constitutive relevance in terms of *mutual manipulability* was supposed to exclude *background conditions* from being mechanistic components. According to Craver, background conditions are not top-down manipulable. For example, the heart's beating is a background condition of the word-stem completion mechanism—it is necessary for the mechanism to run. Still, it is not a component in the mechanism for word-stem completion because one cannot change the phenomenon (word-stem completion behavior) such that the heart's beating changes (Craver, 2007b, p. 157). Still, background conditions seem to satisfy condition (ii): since they are necessary causes of mechanistic components (assuming that the phenomenon indeed occurs), every intervention into a background condition is a common cause of the background condition and the phenomenon (since the latter is constituted by an effect of the background condition). Therefore, based on fat-handedness approaches, it seems to be impossible to distinguish between background conditions and components.

Fifth, fat-handedness approaches (similar to the original mutual manipulability account) represent the phenomenon by one variable only. As a consequence, fat-handedness approaches do not respect the fact that phenomena are processes that involve changes over time (see Kaiser and Krickel (2016); for a modeling account that integrates the temporal dimension see Gebharder and Schurz (2016)). Different components of the mechanism are crucial for different features of the phenomenon at different times. Take the action potential. The action potential is characterized by its rising rate, peak magnitude, declining rate and its refractory period. These features are temporally ordered: first, the potential rises, then it peaks, then it declines, then the refractory period is reached. Explaining the action potential means to account for this temporal order. Only some components are crucial for the rising phase, or the peak magnitude. By representing the phenomenon by only one variable, one cannot account for the fact that changes in the phenomenon map onto changes in the mechanism that occur at different times. Thereby, these accounts neglect the fact that there might be ideal* interventions into phenomena that change more than one component at different times, that again lead to changes in the phenomenon at different times. In the next section, I will present my own suggestion for how to solve the problems of the mutual manipulability account that starts from this idea.

5.2. Taking mutual manipulability seriously

In the remainder of this paper, I will present a novel solution to the problem of how to define constitutive relevance in terms of interventionism. This approach starts with Craver's original definition of constitutive relevance, but interprets mutual manipulability in causal terms while maintaining that causation and constitution are mutually exclusive relations. The core idea of my approach is that manipulability consists in a causal relation between what I call a *temporal EIO-part* (Krickel, 2017) of the phenomenon and a component of the mechanism. To get there, we first have to take a closer look at the metaphysics of mechanistic constitution.

Craver (2007b) argues that the relata of constitutive relevance and mechanistic constitution are *acting entities*. What are acting entities? First, acting entities are concrete particulars. In some sense, acting entities are events (Kaiser & Krickel, 2016): they involve an entity that is doing something, where this doing takes time. Acting entities are things that are spatially and temporally extended in such a way that they can be divided into different parts.

⁷ I want to thank an anonymous reviewer for highlighting the severe consequences of this feature of fat-handedness approaches.

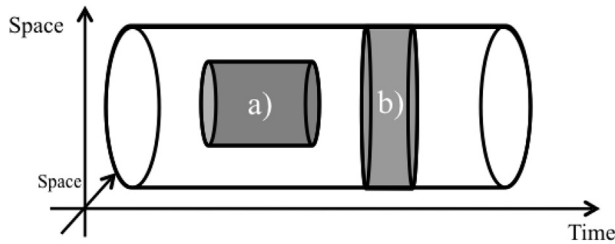


Fig. 5. Illustration of an acting entity and what I call a) a *spatial EIO-part*, and b) a *temporal EIO-part*.

For present purposes, the most relevant kinds of parts are what I have called *spatial EIO-parts* (Fig. 5 a)) and *temporal EIO-parts* (Fig. 5 b)) (Krickel, 2017).⁸ Before I provide definitions of spatial EIO-parts and temporal EIO-parts, I will illustrate the idea with help of an example: the mouse navigating the Morris water maze is an acting entity (the mouse is the entity; the navigation behavior is the activity); its spatial EIO-parts are, for example, the mouse's muscles moving, or the hippocampus generating spatial maps while the mouse is navigating. Its temporal EIO-parts are, for example, the mouse being put into the pool, the mouse swimming for a while in one direction, swimming into another direction, and finally finding the platform.

Temporal EIO-parts are defined as follows:

(*temporal EIO-part*) An acting entity E_1 is a temporal EIO-part of another acting entity E_2 iff:

- (i) the entity involved in E_1 is identical with the entity involved in E_2 ;
- (ii) the activity involved in E_1 begins later and ends earlier than the activity involved in E_2 , or the former begins simultaneously with the latter and ends earlier than the latter, or the former begins later than the latter and ends simultaneously with the latter.

Spatial EIO-parts of acting entities are defined as follows:

(*spatial EIO-part*) An acting entity E_1 is a spatial EIO-part of another acting entity E_2 iff:

- (i) the entity involved in E_1 occupies a proper sub-region of the spatiotemporal region occupied by the entity involved in E_2 ;
- (ii) the activity involved in E_1 occurs during the activity involved in E_2 .

Now, the crucial idea is that acting entities that are spatial EIO-parts and acting entities that are temporal EIO-parts of one and the same acting entity can, in principle, causally interact (if they occur in different space-time regions). This idea is illustrated in Fig. 6.

Consider the mouse navigating the Morris water maze. Assume this navigation behavior lasts from t_1 to t_4 . Now, imagine that during this process the mouse turns left at t_2 (which is a temporal EIO-part of the mouse's navigation behavior). This event might be a cause of the behavior of the hippocampus starting at t_3 (which is a spatial EIO-part of the mouse's navigation behavior). This behavior, again, might be a cause of the mouse's finding the platform at t_4

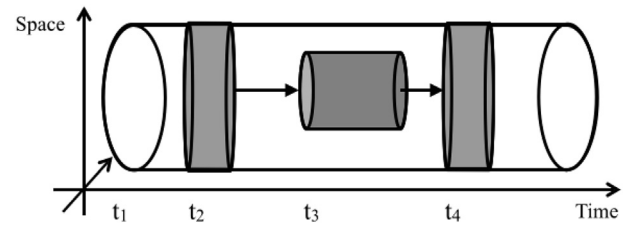


Fig. 6. Causal interactions between spatial and temporal EIO-parts of one and the same acting entity. The x-axis represents time; the y- and z-axis represent spatial dimensions.

(which is a further temporal EIO-part of the mouse's navigation behavior).

Spatial and temporal EIO-parts of one and the same acting entity can in principle causally interact because they do not necessarily occupy the same space-time regions. In other words, they can be wholly distinct events. Remember that the general argument for why events related by constitutive relevance cannot be causally related was that they are not wholly distinct. This objection does not apply here.

Based on these metaphysical considerations, we can solve the problem of how to spell out constitutive relevance in terms of interventions. First, note that since the relata are taken to be concrete individuals, we have to model constitutive relevance in terms of the interventionist framework Woodward developed for *actual causation* that holds between events that actually occurred. According to Woodward's account of actual causation (Woodward, 2003, p. 77), a token event is represented by a variable taking a specific value (written $X = x$). A value of a variable $X = x$ is a (direct) cause of a value of a variable $Y = y$ iff there is an ideal or (ideal*) intervention on X with respect to Y that changes the value of Y given that all other causes of Y that are not on the causal path between I , X , and Y are kept fixed at their actual values (except for variables that I , X , or Y non-causally depend on). Second, since we are talking about temporal EIO-parts of the phenomenon, we have to represent the phenomenon by more than one variable, where each phenomenon-variable represents a temporal EIO-part of the phenomenon. Third, according to what I have argued in Section 4, only if the values of the variables represent wholly distinct events, ideal* interventions can establish causal relations between them.

On the basis of these considerations, what I will call *Causation-based CR* can be defined as follows:

(*Causation-based CR*) X 's Φ -ing is constitutively relevant for S 's Ψ -ing iff.

- (i) X 's Φ -ing is a spatial EIO-part of S 's Ψ -ing,
- (ii) there is a temporal EIO-part of S 's Ψ -ing that is a cause of X 's Φ -ing, and
- (iii) there is a temporal EIO-part of S 's Ψ -ing that it is an effect of X 's Φ -ing.

In order to formulate causation-based CR in terms of interventionism, we have to represent the different acting entities by variables. I will represent X 's Φ -ing by the variable ϕ_i , and the temporal EIO-parts of S 's Ψ -ing by ψ_1 and ψ_2 . Since, as stated above, we are dealing with event/acting entity tokens, we have to apply Woodward's interventionist account of actual causation. In this terminology, condition (ii) is satisfied if and only if there is a variable $\psi_1 = \psi_i$ for which it is true that there is an ideal or ideal* intervention on it with respect to $\phi_i = \phi_i$ that changes ϕ_i while all other variables not on the causal path between ψ_1 and ϕ_i are kept fixed except for variables that ψ_1 and ϕ_i non-causally depend on (Fig. 7,

⁸ "EIO" stands for "entity-involving occurrent," and has been introduced by Marie Kaiser and me (2016) to specify the idea of an *acting entity*. For the purposes of the present paper, the reasons for talking about *entity-involving occurrents* rather than *acting entities* are not relevant. The terms "acting entity" and "entity-involving occurrent" are understood as synonyms in the present context.

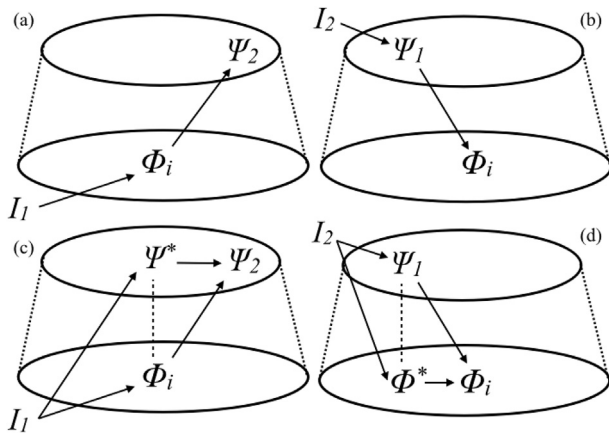


Fig. 7. Causation-based CR in terms of interventionism requires either two ideal interventions (a) (bottom-up intervention) and (b) (top-down intervention), or two ideal* interventions (c) (bottom-up intervention) and (d) (top-down intervention).

(b) and (d)). Condition (iii) is satisfied if and only if there is a variable $\Psi_2 = \psi_2$ for which it is true that there is an ideal or ideal* intervention on $\Phi_i = \phi_i$ that changes Ψ_2 while all other variables not on the causal path between Φ_i and Ψ_2 are kept fixed except for variables that Ψ_2 and Φ_i non-causally depend on (Fig. 7, (a) and (c)). It is plausible to assume that the temporal EIO-parts of a phenomenon constitutively depend on the underlying parts of the respective mechanism. Therefore, interventions into the temporal EIO-parts of the phenomenon are plausibly ideal* (Fig. 7, (c) and (d)).

The models depicted in Fig. 7 allow for a straightforward interpretation of top-down and bottom-up interventions (in interventionist terms and in metaphysical terms) which was a crucial motivation for Craver's mutual manipulability approach. In bottom-up interventions, the variable that is intervened on represents an event/acting entity at the lower level, while the variable with respect to which one intervenes represents an event/an acting entity at the higher level. In the case of top-down interventions it is the other way around.

How does my approach handle the example of the mouse navigating the Morris water maze? Imagine one particular mouse navigating through the maze from t_1 to t_4 . Imagine we want to know whether the hippocampus's activity at t_3 was constitutively relevant to that phenomenon. In order to answer this question, we have to verify, first, whether there was a temporal EIO-part of the mouse's navigation behavior for which it is true that had there been an ideal* intervention on that temporal EIO-part with respect to the hippocampus's activity at t_3 , then the hippocampus's activity at t_3 would have been different. For example, one could imagine that had there been an ideal* intervention on the entering of the mouse into the maze at t_1 (e.g., a change in the location of where the mouse was put into the maze), then the hippocampus's activity would have been different at t_3 (e.g., different neural representations in the hippocampus would have been active). Second, we have to verify whether there was a temporal EIO-part of the mouse's navigation behavior for which it is true that had there been an ideal* intervention on the hippocampus's activity with respect to that temporal EIO-part, the temporal EIO-part would have been different. Again, it seems plausible that had there been an ideal* intervention on the hippocampus's activity at t_3 the mouse's finding the platform at t_4 would have been different (e.g., it would have found the platform at a later time). Hence, my account renders the hippocampus constitutively relevant for the mouse's navigation behavior.

Now assume that the mouse's stomach was active at t_3 , too. The stomach's activity does not seem to be constitutively relevant (assuming a normal mouse and a normal stomach). Plausibly, it is not the case that had there been an ideal* intervention into any temporal EIO-part of the navigation behavior before t_3 with respect to the stomach's activity at t_3 , then the stomach's activity would have been different. Neither does it seem to be plausible that had there been an ideal* intervention into the stomach's activity at t_3 with respect to, for example, the finding of the platform at t_4 , then the finding of the platform at t_4 had been different. Hence, the stomach's digesting is not constitutively relevant for the navigation behavior of the mouse.

There are two crucial challenges for my account: first, one has to tell a story about how scientists can actually test for causation-based constitutive relevance. The problem seems to be that there cannot be interventions into causal processes that already occurred. Second, one has to show how to get to general claims about constitutive relationships from here. Surely, scientists are not so much interested in whether the hippocampus's activity is relevant for the behavior of *that particular* mouse. Rather, they are interested in what constitutes this kind of behavior in general.

Both challenges might be met by adequately comparing sufficiently similar concrete particulars. For example, one compares two instances of a mouse navigating the Morris water maze that differ only with respect to the hippocampus's activity at a particular time. If these instances differ also with respect to the mouse's behavior afterwards, this indicates a causal relation between the hippocampus's activity at that time and the mouse's behavior at that later time. This would correspond to an ideal* bottom-up intervention. Then, one compares two instances of the navigation behavior that differ only with regard to, for example, where the mouse is put into the maze. If this is correlated with differences in the hippocampus's activity during the navigation behavior compared to the other instance, this indicates a causal relation between the first temporal EIO-part of the phenomenon (the mouse being put into the maze at a specific location) and the hippocampus's activity at a later time. In order to reach generalizations, scientists have to perform these experiments with a number of mice (or other animals) in order to exclude the possibility that the effects they observed were due to individual features of the particular mice that have been investigated. For example, there could be a mouse that has a rather weak stomach such that it starts growling every time the mouse moves quickly and that affects the movements depending on how loudly it growls. In this case, the stomach's growling would be constitutively relevant to the moving of that particular mouse. Since not all mice have weak stomachs in this way, the stomach's growling will not come out as constitutively relevant to the navigation behavior of mice in general.

This reinterpretation of Craver's mutual manipulability approach has several advantages over the fat-handedness approach presented in the previous section. First, my approach provides a clear criterion of how to distinguish between interventions that are accidental common causes and interventions that are common causes due to the fact that the variables are related by constitution. If the variables that the intervention is a common cause of are mutually manipulable, then these variables are constitutively related. If they are not mutually manipulable, the intervention is an accidental common cause.

Second, according to my approach, the question whether there is a constitutive relevance relation in a given case is in principle a definite issue. In contrast to fat-handedness approaches, my approach requires only that there are two ideal or ideal* interventions. Since, according to my approach, mutual manipulability is a causal notion, the empirical detectability of constitutive

relevance is only as problematic as the empirical detection of causal relevance.

Third, my approach makes sense of Craver's notions of top-down and bottom-up manipulability. Since fat-handedness approaches assume that mutual manipulability consists in fat-handed interventions, these approaches are unable to make sense of this distinction (see Fig. 7). Craver's notions of top-down and bottom-up interventions are taken to provide a useful and adequate account of the empirical practice of interlevel experiments. It is an advantage of my account that it maintains the descriptive adequacy of Craver's original approach.

Fourth, as argued in the previous section, fat-handedness approaches represent the phenomenon by only one variable. Thereby, they cannot account for the fact that mechanisms are temporally extended and involve changes over time, where these changes map onto different changes in the phenomenon occurring at different times. By representing different temporal EIO-parts of a phenomenon, Causation-based CR can account for the processual character of phenomena and mechanisms.

There might be one final objection: what about components that occur at the same time as the very first temporal EIO-part of the phenomenon, and what about those that occur synchronously with the very last temporal EIO-part of the phenomenon? Condition (ii) cannot be satisfied for the former, since there are no temporal EIO-parts of the phenomenon that occur earlier than the component. Condition (iii) cannot be satisfied for the latter, since there are no temporal EIO-parts of the phenomenon that occur later than the component. One way to solve this problem is to modify the definition of Causation-based CR such that it allows for the first/last component to be such that they satisfy condition (iii)/(ii) and have common causes/common effects with the phenomenon. This alone will not do, at least with regard to the first components that have common causes with the phenomenon, since we run into the problem of accidental common causes that need to be distinguished from common causes that are due to constitutive relevance (see beginning of Section 5.1). Rather, we have to restrict which common causes are allowed in Causation-based CR.

The problem can be solved by invoking the notion of a *fat-handed horizontally surgical (HS) intervention* (Baumgartner, Casini, and Krickel (Under review)). Roughly, a fat-handed intervention I on a phenomenon Ψ and one of its spatial EIO-parts Φ_i is horizontally surgical iff it causes simultaneous changes in Φ_i and Ψ ; if I affects further variables that represent events that are wholly distinct from Φ_i , this influence is mediated via Φ_i ; and I does not affect further variables that represent events that completely overlap with Φ_i (i.e., further properties or activities of the same entity; plausibly, determinables have to be excluded here). A full analysis of this notion cannot be presented here (rather, see Baumgartner, Casini, and Krickel (Under review)). I only want to highlight one important difference between HS interventions and ideal/ideal* interventions as invoked by Causation-based CR. HS interventions require that changes in the phenomenon-variable and the spatial EIO-part-variable occur *at the same time*, while ideal/ideal* interventions require that they change *at different times*. Accounting for both kinds of changes is crucial for the adequacy of an account of constitutive relevance. To see this, take, for example, the phenomenon of a moving car. Two components of the mechanism for the moving of a car, plausibly, are the engine's running and the wheels' turning that both occur while the car is moving (condition (i) of Causation-based CR is satisfied). These two components differ with respect to whether changes in them instantaneously lead to changes in the phenomenon, or whether the changes occur at different times. Plausibly, if the wheels start turning faster, this instantaneously leads to changes in the car's moving (it gets faster). In contrast to that, if the engine changes its

activity, this change has to be transferred to the wheels before it has an impact on the car's velocity. Hence, the change of the engine's activity and the change in the car's moving occur at different times. Therefore, there are no fat-handed horizontally surgical interventions on the engine and the car's moving. Still, the engine's activity satisfies the original definition of Causation-based CR, and thus, turns out to be a component of the car's mechanism.

6. Conclusion

It has been argued that Craver's mutual manipulability account of constitutive relevance is incompatible with interventionism. Constitutive relevance relations do not allow for ideal interventions if one wants to uphold the claim that constitutive relevance is a non-causal relation. I have discussed different strategies for how to solve this problem. Two challenges arose: first, one has to provide a definition of interventionism that does not render constitutive relationships causal. Second, one has to provide a definition of constitutive relevance in terms of interventionism. Different strategies for meeting the first challenge were discussed. The strategy I followed in this paper is to require that the variables in a causal model represent wholly distinct events. Concerning the second challenge, different authors have introduced fat-handedness approaches. I argued that they are problematic. I developed a new suggestion for how to spell out constitutive relevance in terms of interventionism that accounts for Craver's original idea of mutual manipulability. Roughly, mutual manipulability, according to my account, consists in the causal manipulability of the mechanism's components by means of an ideal or ideal* intervention into a temporal EIO-part of the phenomenon, and in the causal manipulability of a temporal EIO-part by means of an ideal or ideal* intervention into the component. As argued, this approach has several advantages over fat-handedness approaches.

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